

Online shopping project

Yes. For **this project**, I would make a GitHub repository specifically showcasing your **B2C e-commerce statistical analysis of Vadodara**.

The project is actually quite useful for your resume because it demonstrates **primary-data collection, sampling, EDA, categorical-data analysis, hypothesis testing, non-parametric methods and R programming**. The report says you collected **323 responses through a 17-question questionnaire** and used R and Excel for computation/processing/visualization.

1. Recommended GitHub repository

Use a professional name such as:

click-conquer-vadodara-ecommerce-analysis

or:

vadodara-ecommerce-consumer-analysis

I prefer the second one for a resume because it immediately tells a recruiter what the project is about.

Structure it like this:

vadodara-ecommerce-consumer-analysis/

|

|— README.md

|

|— data/

|

| |— README.md

|

| |— data_dictionary.csv

|

|— R/

|

| |— 01_data_cleaning.R

|

| |— 02_EDA.R

|

| |— 03_chi_square_tests.R

```
| |— 04_goodman_kruskal_gamma.R
| |— 05_partial_odds_ratio.R
| |— 06_pearson_residuals.R
|
|— results/
| |— figures/
| |— statistical_tables/
|
|— report/
| |— project_report.pdf
|
|— presentation/
| |— presentation.pdf
|
|— README.md
|— requirements.txt
```

Since this project was primarily done using **R + Excel**, I would **not force Python into this repository** unless you actually used Python for the project.

2. Your README should be the main attraction

I'd write something like this:

```
# Vadodara E-Commerce Consumer Analysis
```

```
## Click & Conquer: Unveiling the B2C E-Commerce Odyssey of Vadodara City
```

A statistical study of online shopping behaviour, consumer preferences, platform usage, payment methods, shopping frequency and factors influencing e-commerce decisions among respondents from Vadodara City.

Project Overview

This project investigates B2C e-commerce behaviour in Vadodara using primary survey data collected through a structured questionnaire.

The study combines exploratory data analysis, hypothesis testing and categorical data analysis to understand relationships between demographic characteristics and online shopping behaviour.

Dataset

- 323 respondents
- 17-question questionnaire
- Target population: age 18 and above
- Primary survey data
- Convenience sampling combined with area sampling

Objectives

1. Understand the demographic profile of online shoppers.
2. Identify leading e-commerce platforms and product preferences.
3. Examine relationships between demographic factors and shopping behaviour.
4. Analyse associations using Goodman-Kruskal Gamma and categorical data analysis techniques.
5. Develop a SWOT analysis of the Vadodara e-commerce landscape.

Statistical Methods

- Exploratory Data Analysis
- Chi-Square Test of Independence
- Goodman-Kruskal Gamma
- Partial Odds Ratio
- Pearson's Standardized Residuals
- Contingency Table Analysis
- Descriptive Statistics
- SWOT Analysis

Tools

- R
- Microsoft Excel
- Statistical Analysis
- Data Visualization

Key Findings

- Amazon was the most frequently preferred platform across several product categories.
- Product quality and variety were highly important factors in platform selection.
- Most respondents compared alternatives before purchasing.
- Reviews and ratings were widely used before online purchases.
- UPI was the most frequently used payment method.

- Hidden charges were identified as an important checkout barrier.
- Income group showed a significant association with monthly online shopping expenditure.
- Gender showed no significant association with perceived website-design importance.
- Age and shopping frequency were found to be independent in the reported chi-square analysis.

Project Workflow

Primary Survey



Data Cleaning



Exploratory Data Analysis



Hypothesis Testing



Categorical Data Analysis



Interpretation



SWOT Analysis

Project Report

See the `report/` folder for the complete project report.

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The methodology and objectives above are directly based on your report.

3. Add a project summary at the top

I'd make the first part visually attractive:

 Vadodara E-Commerce Consumer Analysis

—

323 Respondents | 17 Questions | R | Excel | Statistical Analysis

Consumer Behaviour • E-Commerce • Categorical Data • Hypothesis Testing

Then perhaps badges:

R | Statistics | Data Analysis | E-Commerce | Survey Analysis

4. Add a "Project at a Glance" table

This is excellent for recruiters.

Component	Details
Study area	Vadodara City
Data type	Primary
Sample size	323
Questionnaire	17 questions

Component	Details
Target age	18+
Sampling	Convenience + Area sampling
Software	R, Excel
Main analysis	Categorical data
Statistical tests	Chi-square, GK Gamma
Advanced analysis	Partial Odds Ratio, Pearson Residuals
Business analysis	SWOT

Your report explicitly describes the 323 responses, 17 questions and sampling procedure.

5. Put the statistical analysis front and center

This is what differentiates this project from a normal "survey project."

Create:

results/

```
└─ statistical_tables/
    ├── chi_square_results.csv
    ├── gamma_results.csv
    ├── odds_ratio_results.csv
    └─ residual_results.csv
```

And show a summary in the README.

Chi-square analysis

Relationship	χ^2	df	Result
Age × Shopping Frequency	3.0901	5	Not significant
Education × Shopping Frequency	8.6313	6	Not significant
Income × Shopping Frequency	3.1626	5	Not significant

These are the reported values from your project.

6. Showcase Goodman-Kruskal Gamma

This is probably the **most interesting statistical technique** in this project.

Your repository should have a notebook/script:

R/04_goodman_kruskal_gamma.R

And explain:

Goodman-Kruskal Gamma

↓

Ordinal variables

↓

Measure direction + strength

↓

Gamma $\in [-1, +1]$

↓

p-value

↓

Interpretation

Your report explains Gamma as a measure of the strength and direction of association between ordinal variables.

7. Showcase your Partial Odds Ratio

This is another thing I'd emphasize because it demonstrates **categorical-data analysis beyond basic chi-square**.

Create:

R/05_partial_odds_ratio.R

Then show an output table such as:

Age group Female Male Odds Ratio

18–24	13/12	9/15	1.80
25–34	18/30	31/47	0.90
35–44	20/40	18/21	0.58
45–54	4/9	4/7	0.77
55–64	4/8	3/7	1.16
65+	0/1	0/2	1.66

These are from the report's partial contingency analysis.

That's a **very good thing to discuss in an interview**:

"I used age group as a stratification variable and examined gender differences in shopping frequency using partial odds ratios."

8. Add Pearson standardized residuals

Create:

R/06_pearson_residuals.R

And generate a heatmap.

For example:

Occupation × Shopping Frequency

	Frequent	Occasional
Business	+	-
Government	+	-
Homemaker	-	+
Private	+	-
Student	-	+

Your report uses Pearson standardized residuals to examine the difference between observed and expected cell frequencies.

A **residual heatmap** would look much better on GitHub than a screenshot of your report table.

9. Add your strongest visualizations

I'd extract/recreate the best graphs from your report.

Put them here:

results/figures/

Recommended:

01_gender_distribution.png

02_age_distribution.png

03_education_distribution.png

04_occupation_distribution.png

05_shopping_frequency.png

06_platform_preference.png

07_product_category_platform.png

08_purchase_factors.png

09_reviews_usage.png

10_feedback_preferences.png

11_checkout_barriers.png

12_payment_methods.png

13_monthly_spending.png

14_chi_square_heatmap.png

15_gamma_association.png

16_residual_heatmap.png

17_partial_odds_ratio.png

Your report already has visual analyses of platform preferences, shopping frequency, reviews, product feedback, checkout barriers, payment methods and monthly spending.

10. The BEST charts to put in README

Don't put 17 charts into README.

Choose around 5.

I'd choose:

1. E-commerce platform preference

Your report finds Amazon as the leading choice across several product categories.

2. Shopping frequency

Your survey shows occasional shopping as the dominant category.

3. Payment methods

UPI is the leading payment method in the reported results.

4. Checkout barriers

Hidden charges are particularly prominent in the reported responses.

5. Statistical association heatmap

This is the one that demonstrates your **Statistics skill**, rather than just your visualization skill.

11. Add a "Statistical Findings" section

This would make the GitHub much stronger.

For example:

```
## Statistical Findings
```

```
### Age and Shopping Frequency
```

Chi-square test:

$$\chi^2 = 3.0901$$

$$df = 5$$

The reported analysis did not find sufficient evidence of an association between age group and shopping frequency at the 5% significance level.

Income and Monthly Spending

Goodman-Kruskal Gamma:

Gamma = 0.23993

p-value = 0.0001474

The analysis reported a statistically significant positive association between income group and monthly online shopping expenditure.

Your report gives these exact values.

One correction I'd make when you write the GitHub README: your report's interpretation on page 28 says "association between gender and monthly spending," but the tested variable shown is **Income Group × Monthly Spending**. For GitHub, describe it as *income group vs monthly spending*, because that matches the actual contingency table and Gamma output.

12. Add a "Business Insights" section

This is especially useful for **Data Analyst jobs**.

Based strictly on your project, you could write:

Business Insights

- **Platform strategy:** Amazon, Flipkart and Meesho had strong local presence in the survey.
- **Product strategy:** Product quality and variety were important purchase factors.
- **Trust:** Reviews and ratings played an important role in purchase decisions.
- **Payments:** UPI showed strong adoption.
- **Checkout:** Hidden charges and payment-related friction were important barriers.

- **Personalization:** Income-related differences in monthly spending suggest opportunities for differentiated offers.
- **UX:** Website design was generally considered important, but the reported Gamma analysis did not find a significant gender association.

The report's SWOT section also identifies opportunities around localized campaigns, checkout UX, product transparency and vernacular-language/voice-search integration.

13. Add the SWOT analysis

Don't just put a screenshot.

Make it a Markdown table:

Strengths	Weaknesses
Digitally active demographics	Digital divide
Strong platform penetration	Poor/overloaded web design
Reviews influence decisions	Misleading product descriptions
Growing online-shopping trust	Dependence on ratings
Opportunities	Threats
Localized campaigns	Platform saturation
Better checkout UX	Cybersecurity/scams
AR/360° product views	Negative reviews
Gujarati/Hindi integration	Delivery delays

This comes directly from your project's SWOT analysis.

14. Add R code

Since your report explicitly includes R programming, this should be visible.

For example:

R/

|

- |— data_cleaning.R
- |— exploratory_analysis.R
- |— chi_square_analysis.R
- |— gamma_analysis.R
- |— partial_odds_ratio.R
- └─ pearson_residuals.R

Don't dump all the code into one giant file called:

finalcode.R

Separate it logically.

15. Add the report

Yes:

report/

- └─ Click_Conquer_Vadodara_Ecommerce_Report.pdf

The original report is 42 pages and covers the introduction, methodology, EDA, hypothesis testing, categorical analysis, SWOT and references/codes.

But again:

GitHub ≠ PDF storage.

The code + analysis + visuals should be the main attraction.

16. Add data responsibly

Your dataset is based on **323 individual survey responses**.

So I recommend:

data/

- |— README.md

- └─ data_dictionary.csv

If you're allowed to distribute an anonymized dataset:

data/

└─ cleaned_anonymized_survey.csv

If not, don't upload the raw data.

Your data/README.md can say:

Data Availability

The project uses primary survey data collected from respondents in Vadodara City.

The complete raw dataset is not publicly distributed because it contains individual survey responses.

The repository therefore provides the variable definitions and analysis code without exposing potentially sensitive respondent data.

17. One thing I'd improve compared with your original report

Your report uses a lot of **3D/pie charts**.

For GitHub and job applications, I would recreate the important charts using:

- Bar charts
- Horizontal bar charts
- Stacked bar charts
- Heatmaps
- Mosaic plots
- Association plots

rather than simply screenshotting the original Excel charts.

That would make your GitHub look much more like a **professional data-analysis portfolio project**.

18. Your final GitHub structure

I'd ultimately make it:

 vadodara-ecommerce-consumer-analysis

|

|—  README.md 

|

|—  data

|

|— README.md

|

|— data_dictionary.csv

|

|—  R

|

|— 01_data_cleaning.R

|

|— 02_EDA.R

|

|— 03_chi_square.R

|

|— 04_goodman_kruskal_gamma.R

|

|— 05_partial_odds_ratio.R

|

|— 06_pearson_residuals.R

|

|—  results

|

|— figures

|

|— platform_preference.png

|

|— shopping_frequency.png

|

|— payment_methods.png

|

|— checkout_barriers.png

|

|— residual_heatmap.png

|

|

|— tables

|

|— chi_square_results.csv

```
|   ├── gamma_results.csv
|   └── odds_ratios.csv
|
|   ├── 📁 report
|   └── project_report.pdf
|
|   ├── 📁 presentation
|   └── presentation.pdf
|
|   └── 📄 LICENSE
```

★ For your resume

I would link directly to this GitHub repository beside the project:

Click & Conquer: B2C E-Commerce Analysis of Vadodara | R, Excel, Statistical Analysis, Categorical Data Analysis

- Analyzed **323 survey responses** to study e-commerce preferences, shopping behaviour, platform usage and payment patterns.
- Applied **Chi-square, Goodman-Kruskal Gamma, Partial Odds Ratios and Pearson standardized residuals** to evaluate demographic and behavioural associations.
- Developed visual and statistical insights into platform preferences, reviews, checkout barriers and consumer spending patterns.

That is substantially stronger for a **Statistics/Data Analyst resume** than simply writing "E-commerce project using R."